

Myopic Sampling Can Be Maximally Suboptimal in Risk-Limiting Financial Audits

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Abstract

Risk-limiting financial audits estimate a weighted finite-population total by adaptive sampling without replacement. Shekhar, Xu, Lipton, Liang, and Ramdas derive a one-step oracle that samples transaction (i) proportionally to its monetary misstatement contribution $\pi_i f_i$, while leaving the multistage optimal-policy problem open. We show that repeating this oracle can have the worst possible finite-horizon approximation ratio throughout a broad class of confidence sequences. For every $N \geq 2$, risk limit $\delta \in (0, 1)$, and finite uniform betting-stake cap, there are rational weights, taints, and tolerances having optimal expected audit length 1 and oracle expectation

$$1 + \frac{N - 1}{1 + \rho},$$

which approaches N as $\rho \downarrow 0$, for every predictable valid non-control-variate betting strategy obeying the cap. The proof uses the continuous confidence set and does not depend on Approx-Kelly initialization or a candidate grid. If review costs vary by a factor at most κ , the sharp worst-case cost-ratio supremum is $1 + (N - 1)\kappa$; without a bound on cost heterogeneity, no finite guarantee exists even for $N = 2$.

We then give a positive, implementable result for certified AI uncertainty. If simultaneous intervals $\ell_i \leq f_i \leq u_i$ are available, auditing in descending order of $\pi_i(u_i - \ell_i)$ is path-wise minimax-optimal for the resulting robust certificate, for every finite population. We give the matching lower bound, an exact heterogeneous-cost dynamic program, a valid risk-budget composition with betting confidence sequences, exact finite-grid Bellman verification, and reproducible synthetic stress tests. On one family, point scores equal the realized taints yet randomized score sampling approaches both sharp lower bounds, while valid simultaneous intervals identify a one-review, unit-cost optimum. Prediction accuracy and certified audit value are therefore mathematically different.

1 Introduction

Let N transactions have normalized recorded-value weights $\pi_i > 0$, $\sum_i \pi_i = 1$, and unknown misstatement fractions $f_i \in [0, 1]$. The target is the total monetary misstatement

$$m^* = \sum_{i=1}^N \pi_i f_i.$$

At each round an auditor chooses a predictable distribution q_t on the remaining transactions, observes one f_i , updates an anytime-valid confidence sequence, and stops when its diameter is at most ε . Shekhar et al. [1] introduced this risk-limiting financial audit (RLFA) framework. Their

Proposition 2 minimizes the next importance-weighted observation’s variance and maximizes a lower bound on one-step wealth growth using

$$q_t^{\text{oracle}}(i) = \frac{\pi_i f_i}{\sum_{j \in R_t} \pi_j f_j}. \quad (1)$$

They explicitly describe the actual fastest-reduction problem as a difficult multistage optimization and leave a fuller characterization for future work. The issue is consequential: the confidence sequence also contains a logical interval, whose width is the remaining recorded-value mass. Thus a draw can have small local variance value but large terminal value, or vice versa.

This paper makes five contributions.

1. We first solve the released construction exactly for $N = 2$, yielding a transparent counterexample to global oracle optimality.
2. We prove a sharp arbitrary- N theorem for every risk limit and every uniformly bounded predictable betting strategy in the original non-control-variate class. The oracle’s review-count ratio supremum is N .
3. We prove the exact heterogeneous-cost analogue: with cost ratio at most κ , the sharp supremum is $1 + (N - 1)\kappa$; without that restriction the ratio is unbounded already for two transactions.
4. We solve an implementable certified-score problem. For simultaneous AI intervals, descending dollar-weighted interval width is globally optimal, a covering-knapsack dynamic program is exact for heterogeneous costs, and an exact family separates point accuracy from certified audit value.
5. We provide an exact-rational Bellman solver on finite candidate and action grids, independent JSON certificates, exhaustive small-instance checks, and reproducible $N = 1000$ stress tests.

The negative theorems concern stopping time and accumulated review cost; they do not contradict Proposition 2. The bounded-betting result does not cover unbounded stakes or arbitrary control-variate wealth processes. The positive theorem is exact for a robust interval certificate; it does not purport to solve the unrestricted continuous-action betting-CS control problem.

2 Confidence construction and effort objective

For a candidate total m , the betting process updates

$$W_t(m) = W_{t-1}(m)\{1 + \lambda_t(m)(Z_t - \mu_t(m))\}, \quad W_0(m) = 1, \quad (2)$$

where

$$Z_t = \frac{\pi_{I_t} f_{I_t}}{q_t(I_t)}, \quad \mu_t(m) = m - \sum_{s < t} \pi_{I_s} f_{I_s}.$$

The predictable stake must satisfy the usual constraint making each wealth factor nonnegative. Candidates with $W_t(m) \geq 1/\delta$ are rejected. We intersect the betting set with its running intersection and the logical interval

$$[L_t, U_t] = \left[\sum_{s \leq t} \pi_{I_s} f_{I_s}, \sum_{s \leq t} \pi_{I_s} f_{I_s} + \sum_{i \in R_{t+1}} \pi_i \right]. \quad (3)$$

Our main lower bound permits any valid predictable stake-selection strategy for (2) having a finite uniform cap

$$|\lambda_t(m)| \leq L < \infty \quad (4)$$

at every history, time, and candidate. It uses the continuous theoretical set of candidates. The released ApproxKelly implementation [2] is a corollary: it initializes its previous-payoff array at zero, so $\lambda_1 = 0$, and caps $|\lambda_t(m)| \leq 5/2$. The exact two-item result below uses that zero first bet; the arbitrary-population result does not.

The mathematical results use the exact denominator in Z_t . Tiny floating-point stabilizers in the research code are numerical implementation details, not part of the martingale definition.

Definition 2 of [1] calls q_t a probability distribution on the remaining set, and Proposition 2 optimizes over its simplex. We report literal-simplex results and separately state what happens under strict full support.

For known positive review costs c_i , define accumulated manual effort by

$$C_\tau = \sum_{t=1}^{\tau} c_{I_t}, \quad V_c^* = \inf_q \mathbb{E}[C_\tau(q)]. \quad (5)$$

Unit costs recover expected audit length. We call an audit nontrivial when its initial confidence-set diameter exceeds ε , so at least one review is required. The original construction with initial set $[0, 1]$ is nontrivial whenever $\varepsilon < 1$.

3 The exact two-transaction problem

The zero first bet makes the first-round betting set all of $[0, 1]$. After sampling item i , the combined interval therefore has diameter $1 - \pi_i$. Define

$$A(\pi, \varepsilon) = \{i \in \{1, 2\} : 1 - \pi_i \leq \varepsilon\}.$$

Proposition 1 (Complete $N = 2$ solution). *For every first action q ,*

$$\mathbb{E}_q[\tau] = 2 - \sum_{i \in A(\pi, \varepsilon)} q(i). \quad (6)$$

Hence the literal-simplex optimum is 2 if $\varepsilon < \min\{\pi_1, \pi_2\}$, and 1 otherwise. If $A \neq \emptyset$, the minimizers are exactly the distributions with $q(A) = 1$.

Proof. The first draw stops exactly on A . Otherwise the sole remaining item is audited at time two and (3) is a singleton. Thus $\mathbb{E}_q[\tau] = q(A) + 2\{1 - q(A)\}$, and optimizing gives the claim. $\square$

For

$$(\pi_1, \pi_2) = \left(\frac{3}{4}, \frac{1}{4}\right), \quad (f_1, f_2) = \left(\frac{1}{3}, 1\right), \quad \varepsilon = \frac{1}{3}, \quad \delta = \frac{1}{20},$$

the optimum, prop-M, and oracle expectations are respectively

$$1 < \frac{5}{4} < \frac{3}{2}. \quad (7)$$

Under strict full support the infimum remains one but is not attained; putting probability $1 - \eta$ on item 1 gives expectation $1 + \eta$.

4 Sharp bounds under bounded betting

Fix $N \geq 2$, $\delta \in (0, 1)$, and a finite $L \geq 0$. Consider any predictable valid non-control-variate betting strategy satisfying (4). Choose a positive rational ε so that

$$\varepsilon < \frac{1}{3}, \quad (1 + 2L\varepsilon)^N < \frac{1}{\delta}. \quad (8)$$

Such a rational always exists. For any $\rho \in (0, 1]$, set

$$a = \frac{\varepsilon}{2}, \quad w = \frac{a}{N-1}, \quad (9)$$

and define one large transaction and $N-1$ small transactions by

$$\pi_0 = 1 - a, \quad f_0 = \frac{\rho w}{1 - a}; \quad \pi_i = w, \quad f_i = 1 \quad (1 \leq i < N). \quad (10)$$

The contributions are $\rho w, w, \dots, w$, all strictly positive.

Lemma 1 (Two surviving continuous-set witnesses). *Until transaction 0 is audited, the combined running confidence set contains*

$$m_- = a, \quad m_+ = 5a, \quad (11)$$

whose separation is 2ε , for every betting strategy in the stated class.

Proof. If k small transactions have been audited, the logical lower endpoint is $kw \leq a$, while its upper endpoint is at least $\pi_0 > 5a$; the latter follows from $\varepsilon < 1/3$. Thus both witnesses survive (3).

Under (1), write C_R for the remaining contribution and A for the observed contribution. For every possible next item,

$$Z_t = \frac{\pi_i f_i}{\pi_i f_i / C_R} = C_R, \quad Z_t - \mu_t(m) = m^* - m, \quad (12)$$

because $C_R = m^* - A$. Here $m^* = a + \rho w$, so at either witness $|m^* - m| < 2\varepsilon$. Every nonnegative wealth factor is therefore at most $1 + 2L\varepsilon$, regardless of how the predictable stake was initialized or selected. For all $t \leq N$,

$$W_t(m_{\pm}) \leq (1 + 2L\varepsilon)^t \leq (1 + 2L\varepsilon)^N < \frac{1}{\delta}. \quad (13)$$

Neither witness is rejected. Since both survive at every preceding time, the running intersection retains both. $\square$

Theorem 1 (Sharp bounded-betting review-count ratio). *For (9)–(10),*

$$V^* = 1, \quad \mathbb{E}[\tau_{\text{oracle}}] = 1 + \frac{N-1}{1+\rho}. \quad (14)$$

For every fixed $\delta \in (0, 1)$, finite cap L , and bounded predictable strategy in the stated class, the worst-case oracle-to-optimal expected-review ratio supremum over N -transaction instances is exactly N .

Proof. Before item 0 is sampled, the lemma makes the combined diameter at least $2\varepsilon > \varepsilon$. Once item 0 is sampled, the remaining recorded-value weight is at most $a \leq \varepsilon$, so (3) stops the audit. Thus τ_{oracle} is exactly the rank of item 0.

Sequential probability-proportional sampling is the order of independent exponential clocks with the corresponding rates. Each small rate- w clock precedes the rate- ρw clock with probability $1/(1 + \rho)$. Linearity of expectation gives (14). Auditing item 0 first stops in one round by the logical interval, for every betting strategy under consideration. Finally, every nontrivial audit uses at least one and at most N reviews, while (14) approaches N as $\rho \downarrow 0$. $\square$

Theorem 2 (Sharp heterogeneous-cost ratio). *Fix $\kappa \geq 1$. Among nontrivial N -transaction audits in the preceding bounded-betting class with $\max_i c_i / \min_i c_i \leq \kappa$, the worst-case oracle-to-optimal expected review-cost ratio supremum is*

$$\boxed{1 + (N - 1)\kappa}. \quad (15)$$

Without a bound on cost heterogeneity, no finite approximation guarantee exists, even for $N = 2$.

Proof. In (10), assign $c_0 = 1$ and $c_i = \kappa$ for $i \geq 1$. The audit stops exactly when item 0 appears, and each small item precedes it with probability $1/(1 + \rho)$. Hence

$$V_c^* = 1, \quad \mathbb{E}[C_{\text{oracle}}] = 1 + \frac{(N - 1)\kappa}{1 + \rho}, \quad (16)$$

which approaches (15) as $\rho \downarrow 0$.

For the matching upper bound, normalize $\min_i c_i = 1$. A complete audit costs at most $\sum_i c_i \leq 1 + (N - 1)\kappa$, while every nontrivial audit costs at least one. If heterogeneity is unrestricted, the $N = 2$ expression $1 + \kappa/(1 + \rho)$ diverges with κ for any fixed $\rho > 0$. $\square$

Every oracle action has strict full support when $\rho > 0$. If competitors must also have strict support, the values one in (14) and (16) are unattained infima: put first-round mass $1 - \eta$ on item 0, use any fully supported continuation, and let $\eta \downarrow 0$.

Released ApproxKelly corollary. Set $\delta = 1/20$, $L = 5/2$, and $\varepsilon = 1/(20N)$. Then

$$(1 + 2L\varepsilon)^N = \left(1 + \frac{1}{4N}\right)^N < e^{1/4} < 2 < 20.$$

The two witnesses in (11) also lie at indices 1 and 5 of the allowed uniform grid with $40N + 1$ points. Thus the released ApproxKelly construction is a corollary, but neither its zero initialization nor its grid drives the general theorem.

Prop-M stops no later than the rank of item 0, whose expectation is

$$1 + \frac{(N - 1)w}{\pi_0 + w} = 1 + \frac{a}{\pi_0 + w} < \frac{1}{\pi_0}, \quad (17)$$

which approaches one. If point scores happen to equal the realized taints, $S_i = f_i$, prop-MS is identical to (1) and inherits both sharp lower bounds. This is a statement about the constructed scores and outcomes, not a claim that the point model has a certified zero-error guarantee.

5 Certified AI intervals: an exact minimax policy

The negative theorems say that point contributions do not encode terminal uncertainty. Suppose instead that a model, prior audit, or calibrated procedure supplies simultaneous bounds

$$\ell_i \leq f_i \leq u_i \quad (1 \leq i \leq N). \quad (18)$$

After auditing a set B , the exact image of this box under the total is

$$C_B^{\text{box}} = \left[\sum_{i \in B} \pi_i f_i + \sum_{i \notin B} \pi_i \ell_i, \sum_{i \in B} \pi_i f_i + \sum_{i \notin B} \pi_i u_i \right]. \quad (19)$$

Both endpoints are attainable because the uncertainty set is a Cartesian product. Define the dollar-weighted uncertainty

$$d_i = \pi_i(u_i - \ell_i), \quad D(B) = \sum_{i \notin B} d_i. \quad (20)$$

The observation changes the location of (19), but not its width $D(B)$.

Theorem 3 (Pathwise minimax box policy). *Let $d_{(1)} \geq \dots \geq d_{(N)}$, and define*

$$k^* = \min \left\{ k \in \{0, \dots, N\} : \sum_{r=k+1}^N d_{(r)} \leq \varepsilon \right\}. \quad (21)$$

Auditing in nonincreasing order of d_i stops (19) in exactly k^ reviews. Moreover, over all adaptive randomized policies,*

$$\inf_q \sup_{f \in \prod_i [\ell_i, u_i]} \mathbb{E}_f[\tau_{\text{box}}(q)] = k^*. \quad (22)$$

The lower bound holds on every sample path.

Proof. The sorted policy leaves the residual in (21). Any history of length $t < k^*$ can remove at most $\sum_{r=1}^t d_{(r)}$. Its residual width therefore exceeds ε , regardless of observations, randomization, or adaptation. No policy can stop earlier on any path, while the sorted construction attains the bound. $\square$

This gives a precise answer to the implementable robust-certificate problem: audit uncertainty, not merely predicted error. For a multiplicative guarantee, write the condition without division as

$$(1 - b)f_i \leq S_i \leq (1 + b)f_i, \quad 0 \leq b < 1. \quad (23)$$

It implies

$$\ell_i = \frac{S_i}{1 + b}, \quad u_i = \min \left\{ 1, \frac{S_i}{1 - b} \right\}. \quad (24)$$

If no upper endpoint clips, then

$$d_i = \pi_i S_i \frac{2b}{1 - b^2}. \quad (25)$$

Thus $\pi_i S_i$ is the correct priority, but it should be used as a descending deterministic priority for this certificate, not as a randomized probability. When (24) clips, the exact priority in (20) can differ even in rank from $\pi_i S_i$.

Corollary 1 (Point accuracy versus certified audit value). *On the population (9)–(10), let the point scores happen to equal the realized taints, $S_i = f_i$, and supply the simultaneous intervals*

$$[\ell_0, u_0] = \left[0, \frac{2\varepsilon}{\pi_0}\right], \quad [\ell_i, u_i] = [0, 1] \quad (1 \leq i < N). \quad (26)$$

The certified policy stops in one review. Randomized prop-MS has

$$\mathbb{E}[\tau_{\text{propMS}}] = 1 + \frac{N-1}{1+\rho}.$$

With $c_0 = 1$ and $c_i = \kappa$, its expected review cost is (16), while the certified cost optimum is one.

Proof. The large upper endpoint is below one because $\varepsilon < 1/3$. The intervals contain the realized taints: in particular, $\pi_0 f_0 = \rho w \leq 2\varepsilon$. Their uncertainty contributions satisfy

$$d_0 = 2\varepsilon, \quad \sum_{i=1}^{N-1} d_i = \sum_{i=1}^{N-1} w = a = \frac{\varepsilon}{2}.$$

No audited set omitting item 0 can stop, because its residual contains $d_0 > \varepsilon$. Auditing item 0 leaves residual $a \leq \varepsilon$, so descending uncertainty stops after one review. Since $S = f$, prop-MS uses the oracle rates and stops exactly at item 0's rank. The review and cost formulas follow from the exponential-clock calculation. $\square$

5.1 Heterogeneous costs

For general positive c_i , the exact robust problem becomes

$$\begin{aligned} \min_{x \in \{0,1\}^N} \quad & \sum_i c_i x_i \\ \text{s.t.} \quad & \sum_i d_i x_i \geq D(\emptyset) - \varepsilon. \end{aligned} \quad (27)$$

The repository implements an exact rational Pareto-frontier dynamic program. After each item, a state records removed width and cost, caps width at the target, and discards a state if another has at least as much removed width at no greater cost. Induction over items proves that every discarded partial solution is dominated and that the least-cost terminal state solves (27).

5.2 Risk accounting and a hybrid audit

Suppose (18) holds simultaneously with probability at least $1 - \delta_{\text{AI}}$. On that event, (19) covers m^* at every time, under any adaptive policy. If a betting CS has simultaneous failure probability δ_{CS} , their running intersection fails with probability at most

$$\delta_{\text{AI}} + \delta_{\text{CS}} \quad (28)$$

by a union bound; independence is unnecessary. A deployable architecture can therefore individually examine a certainty stratum chosen by (20), then run a randomized risk-limiting procedure on the residual population, explicitly allocating the two risk budgets.

PCAOB AS 2315 describes one sample as more efficient when it achieves the same objectives with a smaller sample, calls for consideration of relative cost and effectiveness when choosing a sampling approach, and distinguishes individually examined items from the sampled residual population [9]. Representative residual samples must still give population items an opportunity for selection. This is operational context, not a legal or professional- compliance conclusion.

Table 1: Exact expected stopping times on the stated finite grid/mesh.

Case	Oracle	prop-M	prop-MS	Certified priority	Mesh Bellman
Terminal value, $N = 3$	2.4167	2.1500	2.3158	2.0556	2.0556
Mixed information, $N = 4$	3.8558	3.4488	3.8302	3.7176	3.4213
Clipped score, $N = 4$	3.2008	3.4488	3.1839	3.4213	3.4213

Table 2: Synthetic certified-box workload; not a client-data claim.

Scenario	Sorted optimum	prop-MS	Oracle	prop-M	Uniform
Unclipped, concentrated	347	452.13	452.68	493.41	895.66
Clipped, high risk	412	536.52	531.01	528.11	898.89
Diffuse values	740	819.31	819.41	895.66	900.44
Tight certificate	659	736.39	736.64	770.81	975.57

6 Finite-grid Bellman verification

For general N , the state must contain the ordered history, remaining set, wealth at every candidate m , ApproxKelly payoff accumulators, and current running intersection. If $\Phi(x, q, i)$ is the transition, the exact equation is

$$V(x) = \begin{cases} 0, & x \text{ is in the stopping region,} \\ 1 + \inf_q \sum_{i \in R(x)} q(i)V(\Phi(x, q, i)), & \text{otherwise.} \end{cases} \quad (29)$$

The action changes both transition probability and confidence state. This is the continuation value absent from the one-step surrogate.

We implement (29) with exact rational arithmetic on a finite candidate grid and the strict-support mesh $q_i \in \{1/D, \dots, D/D\}$. Backward induction over the finite horizon gives the global optimum over that finite action set; no simulation enters these values. Table 1 uses a 21-point candidate grid and $D = 6$. Policies not lying on this action mesh can beat its optimum, so the table is verification and structure discovery, not a continuous-action optimality claim.

The first mesh action in the mixed $N = 4$ case is $(1/6, 1/6, 1/2, 1/6)$, concentrating on neither the largest recorded value nor the largest contribution alone. This is numerical evidence for a blended continuation-value index, not yet a theorem.

7 Reproducible stress tests

We generated four deterministic $N = 1000$ synthetic populations from fixed seeds. Each AI interval follows (24); the stopping rule is solely the simultaneous-box width. The sorted optimum is exact. Comparator means use 1,000 independent exponential-race permutations, which exactly generate their probability-proportional-without-replacement orders. Table 2 reports mean reviews; Monte Carlo standard errors for prop-MS range from 0.17 to 0.30 reviews.

The exact policy uses 9.7–23.3% fewer reviews than randomized prop-MS in these scenarios. These are controlled stress tests of the theorem’s mechanism, not estimates of savings in any audit population. Real validation requires authorized data, simultaneous calibration checks, review-cost measurement, and comparison under the applicable audit objective.

8 Related work and novelty boundary

Waudby-Smith and Ramdas [3] developed confidence sequences for sampling without replacement; Shekhar et al. [1] extended this direction to adaptively weighted monetary totals and AI side information. Imberg et al. [4] study machine-learning-assisted active sampling for finite-population inference. Their optimal importance probabilities use predicted means and covariances to minimize expected asymptotic MSE. That work reinforces the importance of predictive uncertainty, but it does not optimize an RLFA stopping time or the exact robust certificate (19).

Kato and Nakagawa [5] study sequential audit sampling with exact finite-population boundary conditions and expected stopping times. Their model is binary deviation-rate auditing under a hypergeometric random order; they explicitly place monetary-unit sampling outside its direct scope. It neither resolves adaptive weighted selection nor the oracle question here.

Active sequential hypothesis testing considers controlled information acquisition [6], including nonuniform observation costs [7]. Those models repeatedly select noisy experiments to decide among hypotheses and establish dynamic-programming or asymptotic guarantees; they do not give the exact finite-population RLFA ratios here. Goerigk et al. [8] study queries under interval uncertainty for robust knapsack, but with a different packing objective and query budget.

Three ten-angle literature searches, including exact-title citation checks, cost-sensitive sequential design, and 2024–2026 work, found no prior statement of the bounded-betting theorem, sharp review-cost ratio, or exact box policy. This is not a novelty guarantee. Before submission, the result should be checked in subscription indexes, sent to the original authors, and independently reviewed by sequential-analysis and audit-methodology experts.

9 Discussion

The sharp lower bound identifies the control-theoretic failure precisely: zero next-step importance variance does not imply fast entry into a stopping region. In the constructed family, a nearly risk-free but very large recorded value has tiny $\pi_i f_i$, so the oracle postpones the one transaction whose review ends the audit.

The cost theorem turns this from a sample-count pathology into an effort guarantee. With bounded heterogeneity the oracle attains the largest ratio any without-replacement audit can have; without such a bound its ratio is unbounded. The obstruction is again terminal value, not inaccurate point prediction.

Certified intervals turn that diagnosis into an exact implementable policy. They expose how much each unaudited transaction can still move the total. Under independent coordinate bounds this uncertainty is modular, which makes descending $\pi_i(u_i - \ell_i)$ pathwise optimal. The result is deliberately conditional on simultaneous calibration. A point model with good average accuracy but no joint coverage guarantee cannot be inserted into (19) while preserving audit risk.

The calibration separation makes the distinction exact on one population. Point scores can equal every realized taint and still induce nearly maximal work when randomized by predicted contribution. Simultaneous intervals over the same transactions encode a different object—remaining certifiable dollar uncertainty—and recover the exact work plan.

Unbounded betting, arbitrary control variates, the continuous-action solution of (29), correlated uncertainty, and field-data validation remain separate problems. They are not needed for the stated sharp results. The immediate next step is adversarial mathematical and audit-methodology review, not expansion of the theorem surface.

Reproducibility

The repository [10] contains exact rational certificates for (7), (14), (16), (26), and the robust example; independent verifiers that do not import the package; exhaustive interval-grid tests; the augmented-state solver; synthetic benchmark generators; and the manuscript source. The sharp certificate’s $N = 100, \rho = 1/1000$ instance has

$$\mathbb{E}\tau_{\text{oracle}} = \frac{9091}{91} \approx 99.9011, \quad \mathbb{E}\tau_{\text{prop-M}} \leq \frac{396001}{395902} \approx 1.00025.$$

With $\kappa = 100$, the same instance has

$$V_c^* = 1, qquad \mathbb{E}C_{\text{oracle}} = \frac{900091}{91} \approx 9891.1099, \quad \text{sup ratio} = 9901.$$

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